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ARTICLES

Indivisibles, Infinitesimals and a Tale of Seventeenth-Century Mathematics

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Imagine revolving an infinitely long region of infinite area around an axis, only to obtain an object of finite volume. Now imagine computing this finite volume by decomposing it into infinitely many two-dimensional objects and summing their areas. This is precisely what Evangelista Torricelli (1608–1647) did, without the advantage of calculus. His reasoning, while highly intuitive, was not rigorous. It sparked imitation in some (including, perhaps, plagiarism) and rebuke from others. The ensuing debate was part of the rich tapestry of mathematics in the seventeenth century.

The seventeenth century is remarkable for the analytic geometry of René Descartes (1596–1650) and Pierre de Fermat (1601–1665), and the calculus of Gottfried Leibniz (1646–1716) and Isaac Newton (1643–1727). These breakthroughs developed from a mathematical culture concerned with problems of measurement: area, volume, time and distance. The mathematicians who worked just prior to the first publications on calculus tackled very difficult problems of measurement without the benefit of coordinate geometry or the machinery of calculus. Their solutions were often non-rigorous. Yet despite the controversy over the admissibility of their techniques, we cannot help but marvel at their elegance.

In the first part of this paper, we consider two problems solved prior to calculus using intuitive but non-rigorous techniques—a volume problem solved by Torricelli, and an area problem solved by Gilles Personne de Roberval (1602–1675). We also consider an objection to their methods raised by Paul Guldin (1577–1643).

While these problems are easily translated into the language of calculus, the beauty of the original techniques can be lost in translation. In the second part of this paper, we attempt to restore the beauty of the techniques by presenting them within the context of a modern, rigorous method. We avoid much of the machinery of calculus by using infinitesimals. More precisely, we rely on the axiomatic development of infinitesimals introduced in the 1960s by Abraham Robinson (1918–1974). We use infinitesimals, but not the calculus results that arise from their development.

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Indivisibles

Finding the area of a region bounded by curves has its roots in the writings of Eudoxus (408–355 BC) and Archimedes (287–212 BC), who employed the method of exhaustion, approximating an irregular region by filling it with more and more small regions of known area. This idea was also applied to find volumes of irregular solids. While Archimedes had great success with this technique, the insight necessary for constructing such a solution made it difficult to imitate. This method of approximation went essentially unchanged until the beginning of the seventeenth century, when Johannes Kepler (1571–1630) used infinitely small geometric quantities to find volumes of solids of revolution. Kepler's success led the Italian mathematician Bonaventura Cavalieri (1598–1647) to present his method of *indivisibles* to compute areas and volumes in 1635. An indivisible lives one dimension below its environment, like a page in a book (if we allow the conceit of two-dimensional pages). In this way, a page is a slice of a book, and enough pages pressed together create the three-dimensional book. If a page were truly two-dimensional, we would not be able to divide it into two thinner pages; it would be an indivisible.

This metaphor takes us to shaky ground, mathematically. Democritus (ca. 460–370 BC), one of the earliest thinkers to propose this idea, warned of its paradoxes. Imagine slicing a sphere, like an orange, into infinitely thin circular slices; is the last slice a circle, or a point? If a point, then what of the slice next to it? If that slice is not a point, then what is its radius? If we reassemble the slices into the original sphere, will the surface look like tiny stairs?

Despite these philosophical concerns, the theory of indivisibles enjoyed some significant traction with mathematicians of the day. A typical argument using indivisibles followed this strategy: Deconstruct a three-dimensional volume into infinitely many two-dimensional slices, deform each slice into a new shape, and then reassemble the altered slices into a new three-dimensional volume that is equivalent to the original. (The same technique applies if we think of an area as being composed of infinitely many line segments.) This method produces the solutions to the two problems we consider in this paper.

Torricelli finds the volume of Gabriel's Horn

We begin with Torricelli, who used the method of indivisibles to find a volume that, upon first inspection, seems to be infinite. The object resembled a trumpet with an infinitely long neck, stretching upward forever. Thus, it has come to be known as Gabriel's Trumpet, after the angel of that name. The trumpet is one of those marvels of mathematics that reminds us how beautiful a simply defined object can be. It is an object formed by revolving a region of infinite area around an axis. The resulting solid has infinite surface area and finite volume.

The trumpet lies within a hyperbola that is revolved around the vertical axis. In FIGURE 1, we see the hyperbola $y = 1/x$ (for $x > 0$) and its mirror image $y = 1/|x|$ (for $x < 0$), which together define the side view profile of the trumpet's neck. Choose any point C along the positive horizontal axis and locate the corresponding point D on the hyperbola; then $CD = 1/OC$. The region rotated about the vertical axis is bounded by the hyperbola $y = 1/x$, the segment CD , and the horizontal axis. Segment CD passes through $C'D'$ upon revolving, so the trumpet is made finite in the horizontal direction. Its neck, however, extends up forever, narrowing toward G , where the angel Gabriel waits for the divine command to play.

Torricelli peeled away two-dimensional layers of the trumpet as we now describe. For each point E between O and C , locate the corresponding point F on the hyperbola.

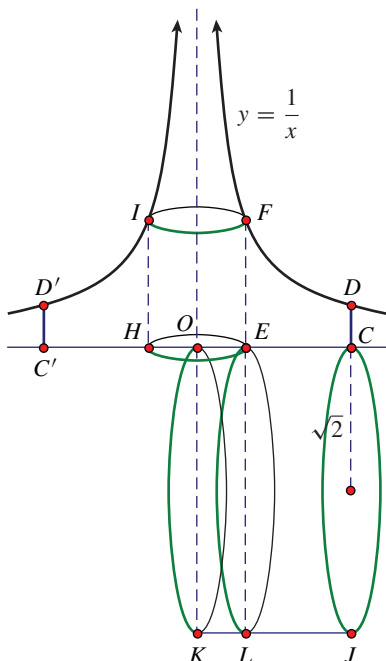


Figure 2 A cylindrical slice of Gabriel's Horn at point E and Torricelli's constructed cylinder

we may prove his result with the cylindrical shell method by calculating an integral of the form

$$2\pi \int_0^C x \cdot \frac{1}{x} dx,$$

a simple task, as long as one is willing to accept the intricate structure supporting calculus.

Roberval calculates the area under a cycloid

When Torricelli was born in 1608, the six-year-old Roberval was learning mathematics in France, over a thousand kilometers away. During their lives, the two men appealed to indivisibles with equal enthusiasm and shared their investigations through regular correspondence. Controversy ensued when Roberval claimed that Torricelli had plagiarized the discovery that we detail in this section; Torricelli died before the matter was resolved, but Roberval's accusation is not supported by modern scholars. Despite the quarrel, the discovery stands as a testament to the elegance of indivisibles.

If the circle with diameter AD in FIGURE 3 were to roll to its right while sitting on AG , the point A would track along the dashed curve AP , peaking at point C , and returning to the "ground" at point G . The curve $APCG$ is called a *cycloid*, and it is generated by a rolling circle; the point A acts like a pebble stuck in the tread of a rolling bicycle tire. The cycloid caught the fancy of mathematicians in the late 1600s, in part because it provided the surprising solution to the following two questions: Along what curve would a clock's pendulum swing so that the clock would keep perfect time no matter how far the pendulum traveled in one swing? What curve allows a ball placed on it to descend from one point to another in the fastest time? In each case, an inverted version of the cycloid shown in FIGURE 3 answers the question.

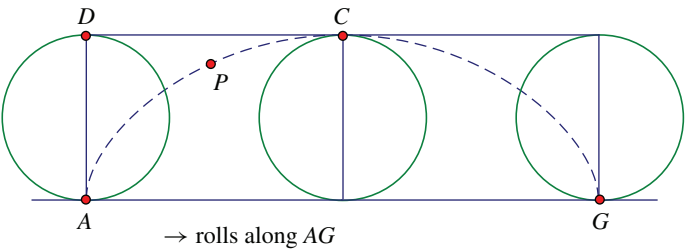


Figure 3 The cycloid $APCG$ tracks the movement of A on the circle as it rolls from A to G

Galileo (1564–1642) approximated the area under a cycloid by building one of metal and weighing it against the metal circle that generated it. This experiment suggested to him that the area of the cycloid is about three times that of the generating circle. Roberval proved that it is *exactly* three times.

As Roberval did, we will find the area under the half-cycloid, and then double this result to find the desired area. FIGURE 4 shows the circle both in its original position and after rolling along half its circumference, forming a half-cycloid. It also shows an intermediate position of the circle, in which point A has moved up to A' , and point E on the circle has become the point of contact labeled E' on the ground AB . Arcs AE and $A'E'$ both share the length of segment AE' , so A has traveled vertically up the same distance that E has traveled vertically down. Construct line $A'E$ to meet AD perpendicularly at L , and place M on the line so that $LE = A'M$. As the circle rolls, this point M traces out a curve just like A does. (Roberval calls AMC the companion curve to the cycloid and finds it to be, using modern symbolism, $y = r - r \cos \frac{x}{r}$, where r is the radius of the generating circle [1, 11].) Since $LE = A'M$ at every moment, Roberval claimed that LE and $A'M$ would sweep through identical areas as point A travels towards point C to form the half-cycloid; hence, the semicircle AED has the same area as region $AA'CM$. This gives us one piece of the area under the cycloid, leaving us to find the area of region $AMCB$.

By symmetry, $AE'ML$ is a rectangle. So the path traveled by M on curve AMC is connected to the path traveled by L on AD . Imagine that the generating circle in FIGURE 4 is animated, and that this circle rolls on AB toward the right at a fixed rate. Point L acts as a reference point that tracks the vertical rise of A as the circle rolls. The ascent of L is not steady: It ascends more and more quickly until the circle has rolled halfway to B , and after that its ascent slows. But while the ascent of L is not steady, it *is* symmetric: The increase of its rate of ascent is mirrored by the decrease as a result of the symmetry of the circle. Therefore, curve AMC divides $ABCD$ in half.

The area of $ABCD$ is its height (the diameter of the circle) times its width (half of the circumference of the circle); that is, the area of $ABCD$ is twice the area of the

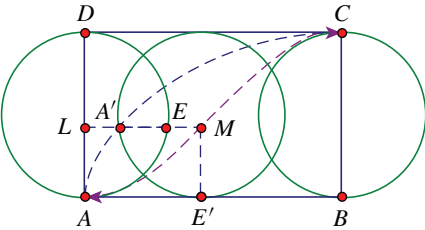


Figure 4 A travels the cycloid while M travels a path that mirrors the height of A

circle. With the area of both pieces resolved, we see the area of half-cycloid $AA'CB$ is half again the area of circle AED . Galileo's experimental estimate was correct: The area under a cycloid is exactly three times the area of the generating circle.

Indivisibles under attack

These discoveries, while celebrated, endured scrutiny by those who believed that mathematical truths should be supported by sound, unambiguous deduction. If a proof contains an approach that can be aimed at a different problem and “prove” an impossibility, then the approach needs repair. The Swiss mathematician Paul Guldin put Cavalieri's method to the test with an example that is as damaging as it is simple to understand [1]. Suppose that triangle $\triangle ADC$, shown in FIGURE 5, has $AD \neq DC$. Altitude BD , therefore, cuts the triangle's base AC into two unequal parts. Guldin proposed that we allow E to vary in its position on AB , as indicated by the arrows. Segment EF is perpendicular to the base, as is GH , and these two segments are equal in length. Thus, to each point E on AB , there corresponds one point H on BC such that $EF = GH$. Guldin, appealing to Cavalieri's method of indivisibles, claimed that if for each segment EF we get a corresponding segment GH , then the areas of triangles $\triangle ADB$ and $\triangle BDC$ must be equal. This absurd conclusion must rest, then, on a spurious claim. Guldin maintained that the only claim that could possibly be spurious in his argument was the one based on Cavalieri's method. Thus, the method is absurd.

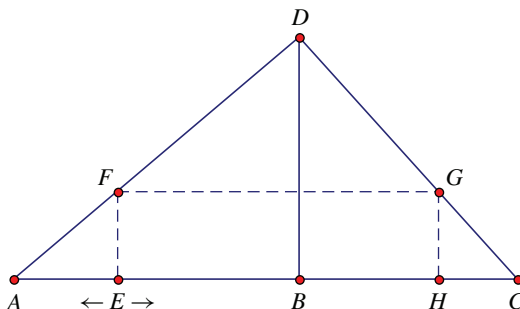


Figure 5 As E slides along segment AB , H mirrors its motion along BC

We shall show why the solutions of Torricelli and Roberval were correct, even though their technique was based on a false premise. It was, in fact, their superb intuition that saved them. With slight modification, rebuilding on the solid foundation of the infinitesimal, we produce the same results, making the work of these mathematicians even more impressive.

Infinitesimal calculus

Torricelli's and Roberval's use of indivisibles gives us a glimpse of the heuristic methods used to solve geometric problems before the invention of the calculus. To be fair, we should remember that the early practitioners of the calculus used infinitesimals in a similarly intuitive way, and lacked a rigorous axiomatic development. Both approaches drew their share of critics. Just as indivisibles faced criticism from Guldin,

infinitesimals drew criticism from George Berkeley (1685–1753). While later mathematicians such as Leonhard Euler (1707–1783) took a step away from a reliance on heuristic arguments, the calculus would have to wait until the nineteenth century for Augustin Louis Cauchy (1789–1857) and Karl Weierstrass (1815–1897) to build a sound logical foundation for the two-centuries-old field. With Cauchy's initial work and Weierstrass's later refinements, calculus took a marked turn away from intuition-based arguments, and infinitesimals were abandoned in favor of limits. Nearly all calculus textbooks and courses still use Weierstrass's ϵ - δ definition of limits as the basis for further study.

While most mathematicians continued to build upon the work of Cauchy and Weierstrass, others tried to formalize the early work of the first practitioners of calculus. Most recently, Abraham Robinson (1918–1974) gave a rigorous development of calculus based on infinitesimals. Robinson gave the name *non-standard analysis* to this flavor of infinitesimal calculus. Every topic in a traditional calculus course can be taught using Robinson's foundation, and some argue that this approach is more natural for students, as it adheres more closely to the ideas that inspired Newton and Leibniz.

The geometric and intuitive solutions given by mathematicians such as Torricelli and Roberval can be revisited in light of Robinson's work. Here we will view their work through this lens. We will introduce only as much of the modern theory as we need. For further study of non-standard analysis, we recommend the original text by Robinson [9], a newer text by Henle and Kleinberg [5], and the first introductory calculus text to use this approach, by Keisler [7].

Non-standard analysis begins with the field of *hyperreals*, denoted by $\mathbb{H}\mathbb{R}$. There are several ways to construct the hyperreals or to prove the existence of such a field, but for our purposes we need only know that the hyperreals form an ordered field in which the reals $\mathbb{R}$ form a designated subfield, and that $\mathbb{H}\mathbb{R}$ contains at least one infinitesimal element. A positive *infinitesimal* is any element of the field that is larger than zero but smaller than any positive real number. We follow the notation of Henle [5] and denote a generic infinitesimal by $\odot$. A negative infinitesimal is smaller than zero but larger than any negative real number. A positive *infinite* hyperreal is larger than any real number, while a negative infinite hyperreal is less than any real number. If $\odot$ is a positive infinitesimal, then $-\odot$ is negative infinitesimal, $1/\odot$ is positive infinite, and $-1/\odot$ is negative infinite. Thus, $\mathbb{H}\mathbb{R}$ contains all of these types of numbers, in addition to the real numbers. A hyperreal is called *nonstandard* if it is not real.

If $\odot \in \mathbb{H}\mathbb{R}$ is a positive infinitesimal, then $\odot < 2\odot < 3\odot < \dots$ and $\odot > \odot^2 > \odot^3 > \dots > 0$. In general, when $0 < x < y$ and y/x is infinite, then x is said to be *infinitely smaller* than y and y is said to be *infinitely larger* than x . Since x/y is infinitesimal, we also say that x is infinitesimal with respect to y . For example, $\odot^2$ is infinitesimal with respect to $\odot$, and $\odot^3$ is infinitesimal with respect to $\odot^2$. (It is interesting to note that John Wallis (1616–1703), the mathematician who introduced the infinity symbol in 1655, not only treated $\frac{1}{\infty}$ as an infinitesimal, but performed arithmetic operations on $\frac{1}{\infty}$ as we do with $\odot$ [10].)

If two hyperreal numbers b and c differ by an infinitesimal or zero, then we say that these numbers are infinitely close, and write $b \approx c$. For example, $4 \approx 4 + \odot$, since $(4 + \odot) - 4 = \odot$. This relation forms an equivalence relation on the hyperreals. Each real number b is surrounded by a cloud of hyperreals that are infinitely close to b . For example, the cloud of hyperreals about 0 are precisely the infinitesimals.

Every finite hyperreal number is infinitely close to a unique real number. The real number that is infinitely close to finite $b \in \mathbb{H}\mathbb{R}$ is called the standard part of b , denoted by $\boxed{b}$; for example, when $\odot$ is an infinitesimal, we have $\boxed{\odot} = 0$ and $\boxed{\pi - \odot^2} = \pi$. (Note that an infinite hyperreal does not have a standard part.) This opera-

tion is a homomorphism that preserves order. In particular, when b and c are finite, we have $\boxed{b+c} = \boxed{b} + \boxed{c}$, $\boxed{bc} = \boxed{b} \boxed{c}$, and $\boxed{b} \leq \boxed{c}$ when $b \leq c$. As another example, when $\odot$ is infinitesimal and r is a real number, $r\odot$ is infinitesimal since $\boxed{r\odot} = \boxed{r} \boxed{\odot} = r \cdot 0 = 0$.

We now return to the work of Roberval, Torricelli, and Guldin, and augment the intuition of indivisibles with the rigor of Robinson's infinitesimals.

Addressing Guldin's objections

Guldin's objections called the validity of the method of indivisibles into question. Now, using infinitesimals, we can address Guldin's concern while remaining faithful to the intuitive method.

Given $\triangle ACD$, place A at the origin and AC along the x -axis. Let B be the orthogonal projection of D onto the x -axis. As shown in FIGURE 6, $A = (0, 0)$, $B = (b, 0)$, $C = (c, 0)$ and $D = (b, h)$, after assigning h as the altitude of $\triangle ACD$. Letting y_1 be the function for the line defined by AD and y_2 be that defined by CD , we have $y_1 = \frac{h}{b}x$ and $y_2 = \frac{h}{b-c}(x - c)$.

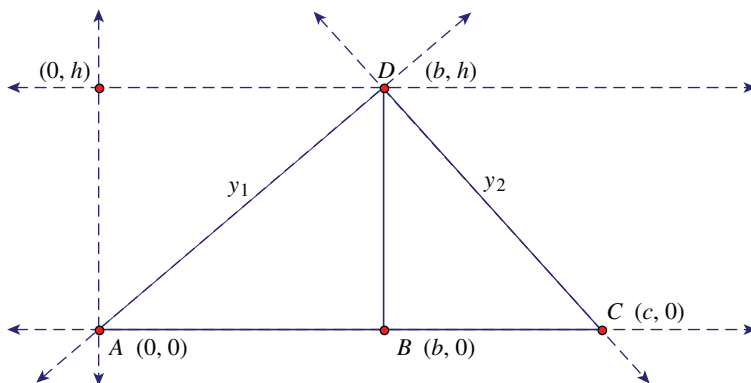


Figure 6 $\triangle ACD$ on the coordinate axes

Let $\odot$ be a positive infinitesimal. We can divide segment AB —that is, the interval $[0, b]$ —into infinitesimal pieces, each of width $b\odot$. Letting $t_\alpha \in [0, b)$ be one of the points of the subdivision, the hyperreals $\{t_\alpha\}$ define an ordered partition of AB . We will designate the subdivision point immediately following a particular t_α , as $t_\beta = t_\alpha + b\odot$.

For every $t_\alpha \in [0, b)$, there exists a corresponding $t'_\alpha \in [b, c)$ which can be determined in the following way, as shown in FIGURE 7. Construct the rectangle with lower left corner t_α , whose upper corners, F and G , lie on $\triangle ACD$. Then, $t'_\alpha \in [b, c)$ is the lower right corner of the rectangle. Using the formulas for y_1 and y_2 , we find that $t'_\alpha = c + \frac{b-c}{b} t_\alpha$.

The hyperreals $\{t'_\alpha\}$ form an ordered partition of BC . There is a bijection of the partitions of AB and BC , but the distance between consecutive elements of the partition of BC is not necessarily the same as that of AB . Since $t_\beta = t_\alpha + b\odot$, using y_1 and y_2 we have

$$t'_\beta = c + \frac{b-c}{b} t_\beta = c + \frac{b-c}{b} (t_\alpha + b\odot).$$

Now we see that the partition of BC has subdivision width $t'_\alpha - t'_\beta = (c - b)\odot$.

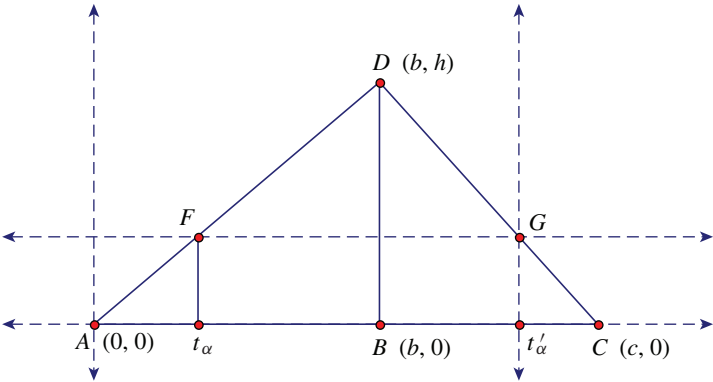


Figure 7 Corresponding partition points t_α and t'_α

Next, let’s compare the infinitesimal areas of the corresponding trapezoids as shaded in FIGURE 8.

$$\begin{aligned} A_L = \text{Area}(\text{left trapezoid}) &= \frac{1}{2}b \odot [y_1(t_\alpha) + y_1(t_\beta)] = \frac{h \odot}{2}[2t_\alpha + b \odot] \\ A_R = \text{Area}(\text{right trapezoid}) &= \frac{1}{2}(c - b) \odot [y_2(t'_\alpha) + y_2(t'_\beta)] \\ &= \frac{h \odot (c - b)}{2b}[2t_\alpha + b \odot]. \end{aligned}$$

Thus, the corresponding individual trapezoidal areas are equal if and only if $b = \frac{c}{2}$. Nevertheless, we must now consider the result of summing these infinitesimal areas. Let Ω_L represent the sum of all left trapezoids A_L , and Ω_R the sum of all right trapezoids A_R . To show that the sums of these left and right infinitesimal areas are not infinitely close when $b \neq c/2$, we will show that $\boxed{\Omega_L - \Omega_R} \neq 0$. First, we note that

$$\Omega_L = \frac{h}{2} \sum_\alpha \odot [2t_\alpha + b \odot] = \frac{h}{2} S,$$

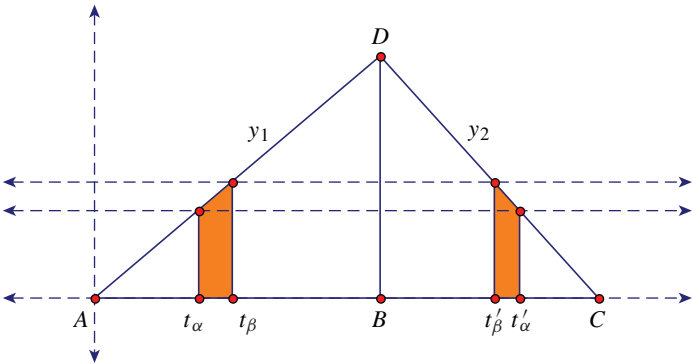


Figure 8 Corresponding trapezoidal areas

where $S = \sum_{\alpha} \odot[2t_{\alpha} + b\odot]$. Similarly,

$$\Omega_R = \frac{h(c-b)}{2b} \sum_{\alpha} \odot[2t_{\alpha} + b\odot] = \frac{h(c-b)}{2b} S.$$

Notice that these are infinite sums of hyperreals, and since $\triangle ABD$ and $\triangle CBD$ have finite positive real area, these sums are neither infinitesimal nor infinite. A formal treatment of series in the context of the hyperreals can be found in Chapter 5 of [6], Chapter 6 of [4], or Chapter 9 of [5]. For our purposes, we need only know that the distributive law applies in order to justify the factoring in these two equations.

Since the area of $\triangle ABD = \boxed{\Omega_L}$ is a positive real number, we know that S is not infinitesimal. Next, subtraction gives

$$\Omega_L - \Omega_R = \frac{h}{2} S - \frac{h(c-b)}{2b} S = \frac{h(2b-c)}{2b} S.$$

Neither of the factors on the right is infinitesimal, so neither is $\Omega_L - \Omega_R$. Hence, $\triangle ABD$ and $\triangle CBD$ are only equivalent in area when $\triangle ACD$ is isosceles, and Guldin's criticism has been addressed.

Roberval's method revisited

Roberval determined the area under the cycloid by finding the area under the half-cycloid OPC in FIGURE 9. Suppose that r is the radius of the generating circle OAD . Recall that, by Roberval's construction, we have: Segment LA is perpendicular to diameter OD , and points L, A, P and M all lie on the same perpendicular line, with M defined such that $LA = PM$. Also, as P travels the half-cycloid from O to C , M 's path traces a curve OMC that divides the rectangle $OBCD$ into congruent pieces. Thus, the area of the shaded region in FIGURE 10 is πr^2 , since OB is half the circumference of the generating circle and OD is its diameter.

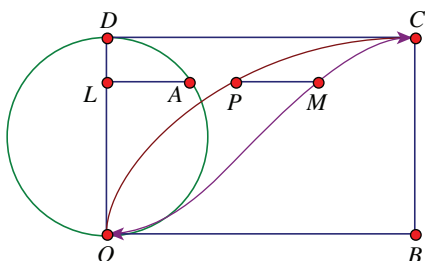


Figure 9 Cycloid with generating circle

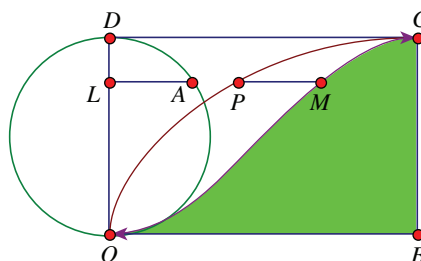


Figure 10 Part 1 of cycloidal area

Roberval then employed the theory of indivisibles to show that the area of the crescent-shaped shaded region in FIGURE 11 is equivalent to the area of semicircle OAD . Let's update his solution by using the hyperreals. Let $\odot$ be a positive infinitesimal. Create a partition of OD consisting of subintervals of equal width $\odot$. Suppose that L and L' are partition points separated by $\odot$. We will consider two regions corresponding to a subinterval of the partition: the region $LL'A'A$ trapped within the generating circle, and $PP'M'M$ trapped between the cycloid OPC and the curve OMC , as shown

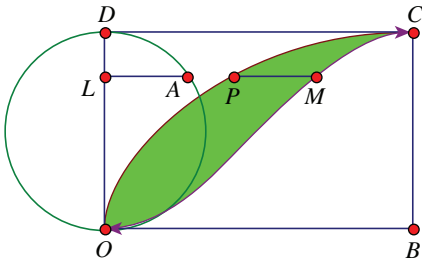


Figure 11 Part 2 of cycloidal area

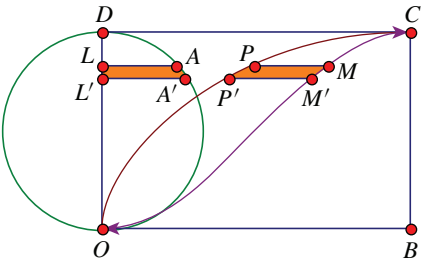


Figure 12 Corresponding regions

in FIGURE 12. Notice that $LA = PM$, $L'A' = P'M'$ and $LL' = \odot$, by construction. We will compare the areas of these corresponding regions.

First, let's consider these regions as they exist in the lower half of the generating circle, as shown in FIGURE 13. In this figure, we have labeled $w = LA = PM$ and $w' = L'A' = P'M'$. Also notice that $w' < w$, since these regions are on the lower half of the generating circle. For Area_1 , it is clear that

$$w' \odot < \text{Area}_1 < w \odot . \tag{1}$$



Figure 13 Corresponding regions in the lower half of rectangle $OBCD$

Within Area_2 , drop a perpendicular from PM that intersects $P'M'$ at Q , as shown in FIGURE 14. By construction we have $PQ = \odot$. Let's consider the length of segment $P'Q$. Recall that P' and P are points on the half-cycloid OPC . Since the cycloid is an increasing function, a non-infinitesimal positive change in input produces a non-infinitesimal positive change in output. Therefore, $P'Q$ is positive infinitesimal, since PQ is positive infinitesimal. Letting $P'Q = \odot_2$, it is clear that

$$(w' - \odot_2) \odot < \text{Area}_2 < (w + \odot_2) \odot . \tag{2}$$

Also, since $\odot_2$ is positive infinitesimal, Inequality (1) gives

$$(w' - \odot_2) \odot < w' \odot < \text{Area}_1 < w \odot < (w + \odot_2) \odot . \tag{3}$$

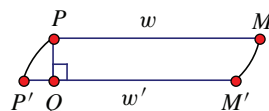


Figure 14 Area_2 bounded by cycloid and curve OMC

To find a bound on $w - w'$, we'll use the fact that A and A' are points on the lower half of the generating circle, as seen in FIGURE 15. The point S is the center of the circle. Thus, $SA = SA' = r$, where r is the radius of the generating circle, and

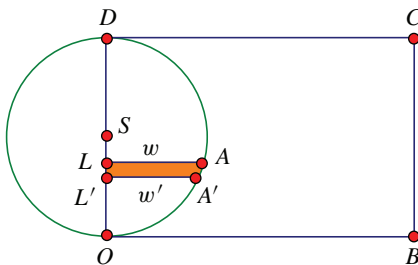


Figure 15 Area₁ within the generating circle

$LL' = \odot$ by construction. Notice that $w^2 = r^2 - (LS)^2$ and $(w')^2 = r^2 - (L'S)^2 = r^2 - (LS + \odot)^2$. Therefore,

$$\begin{aligned} (w - w')^2 &\leq (w - w')(w + w') \\ &= w^2 - (w')^2 \\ &= (r^2 - (LS)^2) - (r^2 - (LS + \odot)^2) \\ &= 2(LS) \odot + \odot^2 \\ &\leq 2r \odot + \odot^2 \text{ since } LS \leq r. \end{aligned}$$

Thus, $|w - w'| \leq \sqrt{2r \odot + \odot^2}$ for all such w and w' . The argument for the upper half of the generating circle is similar.

Let Ω_1 represent the sum of all Area₁ sections (producing the area of semicircle OAD) and Ω_2 the sum of the Area₂ sections (producing the area of region $OPCM$). We will show that the difference between these sums is infinitesimal, hence the real areas must be the same. Inequalities (2) and (3) give

$$|\Omega_1 - \Omega_2| \leq \odot \sum [w - w' + 2\odot_2] = \odot \sum [w - w'] + 2 \sum [\odot \odot_2],$$

where we are summing over all $\frac{2r}{\odot}$ pieces of the partition of OD . For the first summation, our bound on $w - w'$ allows us to simplify as follows:

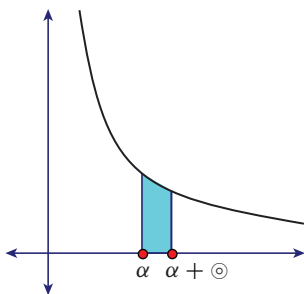
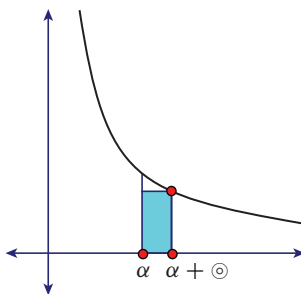
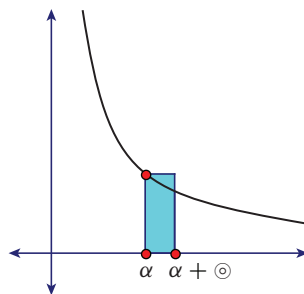
$$\odot \sum [w - w'] \leq \odot \cdot \left[\frac{2r}{\odot} \sqrt{\odot(2r + \odot)} \right] = 2r \sqrt{\odot(2r + \odot)},$$

an infinitesimal quantity. For the second summation, we only need to note that $\odot \odot_2$ is infinitesimal with respect to $\odot$. Since we are summing over $\frac{2r}{\odot}$ pieces of the partition, this sum is at most the same order as $\odot_2$, thus infinitesimal. We can now conclude that $|\Omega_1 - \Omega_2| = 0$, since both sums are infinitesimal.

Thus, the area of crescent $OPCM$ is half the area of the generating circle. Combining this with the area of region $OMCB$, we see that the area of the half cycloid OPC is $\frac{3}{2}\pi r^2$. Therefore, the area of the cycloid is three times that of its generating circle.

Torricelli's method revisited

Let's revisit Gabriel's horn using infinitesimals. Torricelli considered the solid generated by rotating the region under the curve $y = \frac{1}{x}$ on the interval $(0, c]$, where $c > 0$, about the y -axis. Divide the hyperreal x -axis interval $(0, c]$ into segments of equal

**Figure 16** Subinterval area**Figure 17** Lower estimate**Figure 18** Upper estimate

width of positive infinitesimal length $\odot$, and suppose that $x = \alpha$ is one of the subdivision points. Here, $[\alpha, \alpha + \odot]$ is a typical subinterval of the partition.

Rotating the shaded region trapped between the curve and the x -axis interval $[\alpha, \alpha + \odot]$, as shown in FIGURE 16, about the y -axis, produces a solid best described as a piece of pipe with a curved top edge and a wall thickness of $\odot$. The curved top edge makes this volume a bit more difficult to calculate than we would like. Instead, consider the cylindrical shells that form a lower bound and an upper bound for this pipe. Specifically, the volume of this pipe is bigger than the cylindrical shell of wall thickness $\odot$ produced by using the lower height $\frac{1}{\alpha + \odot}$ at the right endpoint of the interval, as shaded in FIGURE 17, but smaller than the shell of thickness $\odot$ produced by using the higher height $\frac{1}{\alpha}$ of the left endpoint, as shaded in FIGURE 18. The volume of an actual piece of pipe with curved top lies between these bounds; that is,

$$\text{Vol}_{\text{right endpoint shell}} \leq \text{Vol}_{\text{pipe}} \leq \text{Vol}_{\text{left endpoint shell}}.$$

This yields

$$\frac{\pi \odot (2\alpha + \odot)}{\alpha + \odot} \leq \text{Vol}_{\text{pipe}} \leq \frac{\pi \odot (2\alpha + \odot)}{\alpha}.$$

Simplification produces

$$2\pi \odot - \frac{\pi \odot^2}{\alpha + \odot} \leq \text{Vol}_{\text{pipe}} \leq 2\pi \odot + \frac{\pi \odot^2}{\alpha}.$$

Since $\alpha + \odot > \alpha$, we have

$$2\pi \odot - \frac{\pi \odot^2}{\alpha} \leq \text{Vol}_{\text{pipe}} \leq 2\pi \odot + \frac{\pi \odot^2}{\alpha}. \quad (4)$$

Thus, the volume of a piece of pipe is within $\frac{\pi \odot^2}{\alpha}$ of $2\pi \odot$.

While Torricelli's solution is more straightforward than Roberval's, at this point his method is more difficult to reconcile with infinitesimals because of the complexity added by the infinite discontinuity at the origin. We'll have to split the argument into two pieces of consideration, infinitesimal values of α and non-infinitesimal values of α .

Let's start by considering the volume produced by the partitions with non-infinitesimal values of α . When α is non-infinitesimal, $\frac{\pi \odot^2}{\alpha}$ is infinitesimal with respect to $2\pi \odot$. That is, $2\pi \odot / (\frac{\pi \odot^2}{\alpha}) = \frac{2\alpha}{\odot}$ is infinite. This allows us to simplify Inequality (4) and say that the volume of each pipe resulting from a non-infinitesimal α lies between $2\pi \odot + \odot'$ and $2\pi \odot - \odot'$, where $\odot'$ is infinitesimal with respect to $\odot$.

Now, we need to sum the volumes from these pipes. We started with $\frac{c}{\odot}$ intervals, but we have excluded all that are infinitely close to 0, that is, those where α is infinitesimal.

Let r be the number of intervals that we have eliminated. Notice that $r\odot$ must be infinitesimal, else it is not infinitely close to 0 and we have excluded a non-infinitesimal α . This gives

$$\sum_{\boxed{\alpha} \neq 0} [2\pi \odot - \odot'] \leq \sum_{\boxed{\alpha} \neq 0} \text{Vol}_{\text{pipe}} \leq \sum_{\boxed{\alpha} \neq 0} [2\pi \odot + \odot'],$$

where $\odot'$ is infinitesimal with respect to $\odot$. Let's consider the two components of the summations on the left and right. Since there are $\frac{c}{\odot} - r$ non-infinitesimal α , the first part of each simplifies to

$$\sum_{\boxed{\alpha} \neq 0} 2\pi \odot = \left(\frac{c}{\odot} - r \right) \cdot 2\pi \odot = 2\pi c - 2\pi \odot r.$$

As noted above, $r\odot$ is infinitesimal. Thus,

$$\boxed{\sum_{\boxed{\alpha} \neq 0} 2\pi \odot} = \boxed{2\pi c} - \boxed{2\pi \odot r} = 2\pi c.$$

For the second contributor to these summations, $\sum \odot'$, we only need to note that $\odot'$ is infinitesimal with respect to $\odot$. Since we are summing over $\frac{c}{\odot} - r$ such α , the sum is at most infinitesimal. Therefore,

$$\boxed{\sum_{\boxed{\alpha} \neq 0} \odot'} = 0,$$

and we have

$$\boxed{\sum_{\boxed{\alpha} \neq 0} \text{Vol}_{\text{pipe}}} = 2\pi c.$$

Now let's consider the volume produced by the partitions where α is infinitesimal. By Inequality (4) we have

$$\begin{aligned} \sum_{\boxed{\alpha} = 0} \text{Vol}_{\text{pipe}} &\leq \sum_{\boxed{\alpha} = 0} 2\pi \odot + \frac{\pi \odot^2}{\alpha} \\ &\leq \sum_{\boxed{\alpha} = 0} 2\pi \odot + \frac{\pi \odot^2}{\odot}, \text{ since each } \alpha \geq \odot, \\ &= \sum_{\boxed{\alpha} = 0} 3\pi \odot \\ &= r(3\pi \odot), \end{aligned}$$

since there are r such infinitesimal α .

Since $r\odot$ is infinitesimal, we have

$$\boxed{\sum_{\boxed{\alpha} = 0} \text{Vol}_{\text{pipe}}} \leq \boxed{3\pi r \odot} = 0.$$

Thus, the infinitesimal values of α contribute no real volume to the horn.

In conclusion, since the volume of the horn is a real number, and we have bounded the standard part of the sum of the volumes of these pipes above and below by $2\pi c$, we have

$$\text{Vol}_{\text{horn}} = \sum_{\alpha} \text{Vol}_{\text{pipe}} = 2\pi c.$$

Conclusion

We have been able to salvage some wonderful intuition applied to interesting problems through the use of infinitesimals. If the reader would like to verify further seventeenth-century “proofs” by indivisibles using infinitesimal calculus, we suggest Margaret Baron’s *The Origins of the Infinitesimal Calculus* [1] as a rich source of such proofs. Baron presents arguments like Roberval’s by Fermat, Descartes and Christiaan Huygens (1629–1695), each of whom determined the area under a cycloid using indivisibles. John Wallis used indivisibles to determine the volume of a cylindrical section on a semicircular base. Luca Valerio (1552–1618), who inspired Cavalieri, found the volume of a hemisphere in a manner that blended indivisibles with the modern concept of limit.

Finally, Baron details many results via indivisibles produced by Kepler, who used the method with abandon. Of him, Baron writes that he “sought to demonstrate in the simplest possible way the existence of mathematical form and structure in the external world and, upheld by his Platonic-Pythagorean philosophy, he often allowed faith, analogy and intuition to guide him when traditional methods failed” [1]. With Robinson’s theory of infinitesimals, the intuition of these prolific mathematicians who expanded the boundaries of seventeenth century mathematics rests on a solid foundation.

REFERENCES

1. M. Baron, *The Origins of the Infinitesimal Calculus*, Dover, New York, 1969.
2. C. Boyer, *The History of the Calculus and Its Conceptual Development*, Dover, New York, 1949.
3. C. H. Edwards, Jr., *The Historical Development of the Calculus*, Springer-Verlag, New York, 1979. <http://dx.doi.org/10.1007/978-1-4612-6230-5>
4. R. Goldblatt, *Lectures on the Hyperreals*, Springer-Verlag, New York, 1998. <http://dx.doi.org/10.1007/978-1-4612-0615-6>
5. J. M. Henle and E. M. Kleinberg, *Infinitesimal Calculus*, Dover, New York, 2003.
6. R. F. Hoskins, *Standard and Nonstandard Analysis*, Ellis Horwood, London, 1990.
7. H. J. Keisler, *Elementary Calculus: An Infinitesimal Approach*, 2nd ed., Prindle, Weber & Schmidt, Boston, 1986. Currently available online at <http://www.math.wisc.edu/~keisler/calc.html>.
8. P. Mancosu, *Philosophy of Mathematics and Mathematical Practice in the Seventeenth Century*, Oxford, New York, 1996.
9. A. Robinson, *Non-standard Analysis*, North-Holland, Amsterdam, 1966.
10. J. Stedall, *The Arithmetic of Infinitesimals: John Wallis 1656*, Springer, New York, 2004.
11. D. Struik, *A Source Book in Mathematics, 1200–1800*, Princeton University Press, Princeton, 1986.

Summary In this article, we describe clever arguments by Torricelli and Roberval that employ indivisibles to find the volume of Gabriel’s trumpet and the area under the cycloid. We detail 17th-century objections to these non-rigorous but highly intuitive techniques, as well as the controversy surrounding indivisibles. After reviewing the fundamentals of infinitesimal calculus and its rigorous footing provided by Robinson in the 1960s, we are able to revisit the 17th-century solutions. In changing from indivisible to infinitesimal-based arguments, we salvage the beautiful intuition found in these works.

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